

Day 19 - Lecture 19.2: Multiple Datasets on Histogram

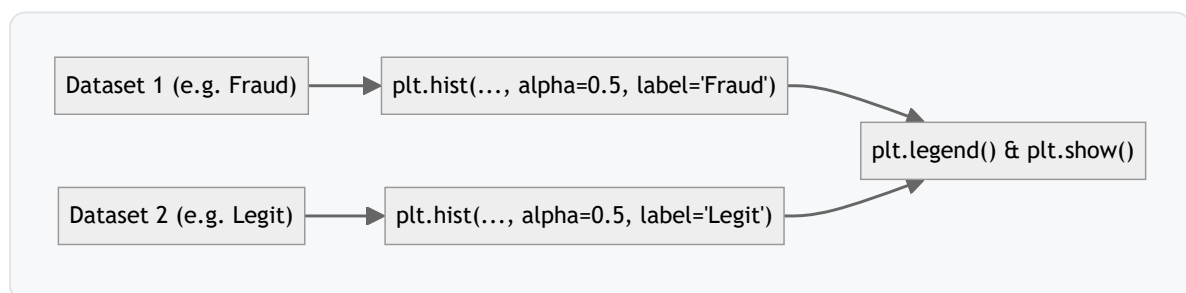
1. Overview & Purpose

In real-world data science, we frequently need to **compare the distributions of two or more populations** (e.g., Male vs. Female heights, Fraudulent vs. Legitimate transactions, Control vs. Treatment groups).

Plotting multiple datasets on the same histogram reveals:

- **Shift in central tendency:** Is one group systematically higher than the other?
- **Differences in variance/spread:** Is one group more dispersed or volatile?
- **Overlap & separability:** Can a classifier easily separate the two classes based on this feature?

2. Core Techniques & Parameters



Key Parameters for Multiple Histograms:

1. **alpha (Transparency):** Set `alpha=0.5` or `0.6` so that overlapping regions between datasets are clearly visible rather than one dataset occluding the other.
2. **label and plt.legend():** Essential for distinguishing which color corresponds to which group.
3. **Aligned Bins:** To make a fair visual comparison, ensure both datasets share either the same number of bins or identical bin edges.

4. Side-by-side vs Overlaid:

- **Overlaid:** Call `plt.hist()` twice on the same axes with `alpha` .
- **Side-by-side:** Pass a list of arrays: `plt.hist([data1, data2], label=['D1', 'D2'])` .

3. Real-World Case Study: Fraud Detection

In credit card transactions:

- **Legitimate transactions:** High volume of small everyday purchases (2—50), with a long tail tapering off.
- **Fraudulent transactions:** Higher average amount, concentrated in round large values (500,1000, \$2000).

```
import matplotlib.pyplot as plt

legit_transactions = [
    2.99, 5.49, 8.99, 12.50, 14.99, 19.99, 23.45, 29.99, 34.99, 39.50,
    45.00, 49.99, 55.25, 60.00, 75.99, 89.99, 120.50, 150.00, 199.99,
    249.99, 300.75, 450.00, 600.00, 850.00, 1200.00
]

fraud_transactions = [
    50, 100, 150, 200, 300, 500, 500, 750, 1000, 1000,
    1200, 1500, 1500, 2000, 2500, 3000, 3000
]

plt.figure(figsize=(10, 6))
plt.hist(fraud_transactions, bins=10, label='Fraudulent', color='crimson', alpha=0.5)
plt.hist(legit_transactions, bins=10, label='Legitimate', color='seagreen', alpha=0.5)

plt.xlabel('Transaction Amount ($)')
plt.ylabel('Frequency')
plt.title('Transaction Amount Distribution: Legitimate vs Fraudulent')
plt.legend(fontsize=11)
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```

4. Key Takeaways & Best Practices

- When data scale varies dramatically (e.g. 1,000,000 legit vs 500 fraud), raw counts make the smaller class invisible. Use `density=True` (probability density) or a log scale (`plt.yscale('log')`).
- Keep colors intuitive: Red/Crimson for anomalies/fraud/errors, Green/Blue for normal/legitimate data.